American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
10	3/27/26	9 am	50 minutes	⦿ Advisor ○ Students	Raghad Al Agha

Attendees:

Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury

Briefly summarize the main discussions during the meeting:

The meeting covered progress on the mobile application, machine-learning model evaluation, and supporting documentation. On the app side, the team finalized key UI pages including welcome, sign-in/sign-up (with Google integration option), scanning, homepage, analytics, history, and settings, while simplifying navigation for usability, especially for elderly users. Additional features such as tutorials, resources, and accessibility improvements (font and guidance systems) are being designed. On the modeling side, the team analyzed inconsistencies in a reference paper, particularly regarding dataset splitting and augmentation practices, and implemented a more reliable pipeline using a 68/17/15 split with k-fold validation. They also introduced Optuna (Bayesian optimization with TPE) for hyperparameter tuning, achieving improved test accuracies around 92–94%, while avoiding overly optimistic results caused by improper augmentation. The discussion also included progress on the app datasheet design (combining user-friendly and technical layers), and clarification of conference submissions, where the team confirmed acceptance of multiple formats (paper, demo, poster) and reviewed feedback emphasizing the need for real-world testing and full system integration.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that the app design is almost complete at the structural level, with Google sign in/up option successfully integrated and usability improvements progressing well. The machine-learning pipeline is now more robust and methodologically correct, with realistic performance metrics achieved through proper data handling and hyperparameter tuning. The datasheet will be developed as a layered document accessible within the app for users. Regarding the conference, the accepted paper will be the primary focus, while poster/demo participation will be clarified with organizers, and efforts will be directed toward addressing feedback, particularly completing system integration and obtaining real-world testing results.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Help in implementing the UI design in the application, work on app tutorial, work on the conference paper.	Raghad Al Agha	Wed, April 1, 2026
Implementing the UI in application and app tutorial, work on the conference paper.	Lynn Ariss	Wed, April 1, 2026
Help in integrating the Backend of the app, search on optimizing the Ataxia ML model, work on the conference paper.	Hadi Elham	Wed, April 1, 2026
Work on scaling down the whole device, work on conference paper	Claritta Khoury	Wed, April 1, 2026